

Guy Wilson

ghwilson@stanford.edu

guyhwilson.com

Education

Stanford University

Ph.D. Student, *Neuroscience*.

University of California, Berkeley

B.A. *Mathematics, Molecular & Cell Biology*, 2018.

Awards and Honors

NSF Graduate Research Fellowship, Stanford, 2018.

Regina Casper Stanford Graduate Fellowship, Stanford, 2018.

Phi Beta Kappa (offered), UC Berkeley, 2018.

Departmental Honors (MCB), UC Berkeley, 2018.

Publications

GH Wilson*, SD Stavisky*, FR Willett, DT Avansino, LR Hochberg, JM Henderson**, S Druckmann**, KV Shenoy**. Decoding spoken English phonemes from intracortical electrode arrays in dorsal pre-central gyrus. *Biorxiv preprint*, 2020.

SD Stavisky, FR Willett, **GH Wilson**, B Murphy, P Rezaii, DT Avansino, WD Memberg, JP Miller, RF Kirsch, LR Hochberg, AB Ajiboye, S Druckmann, KV Shenoy, JM Henderson. Neural ensemble dynamics in dorsal motor cortex during speech in people with paralysis, *eLife*, 2019.

RF Helfrich, M Huang, **GH Wilson**, RT Knight. Prefrontal cortex modulates posterior alpha oscillations during top-down guided visual perception, *Proceedings of the National Academy of Sciences*, 2017.

Posters

Wilson G, Stavisky SD, Avansino D, Hochberg LR, Henderson JM, Shenoy KV, Druckmann S. Decoding spoken English phonemes from dorsal motor cortex. *BCI Society International Meeting, cancelled due to COVID-19*, 2021.

Wilson G, Stavisky SD, Avansino D, Hochberg LR, Henderson JM, Shenoy KV, Druckmann S. Neural state space geometry in human motor cortex underlying speaking different phonemes. *Computational and Systems Neuroscience (COSYNE)*, 2020.

SD Stavisky, FR Willett, **GH Wilson**, P Rezaii, DT Avansino, LR Hochberg, KV Shenoy, JM Henderson. Similar low dimensional neural population dynamics in dorsal motor cortex during human speech and hand movements, *Society for Neuroscience Annual Meeting*, 2019.

GH Wilson, RF Helfrich, RT Knight. Spatiotemporal Dynamics of Contextual Processing in Human Intracranial Recordings, *MCB Honors Research Symposium*, 2018.

M Huang, RF Helfrich, **GH Wilson**, RT Knight. Sequential and Temporal Predictions in the Human Brain Employ Neuronal Alpha Oscillations, *MCB Honors Research Symposium*, 2017.

Talks

B Sheffer and **GH Wilson**. Linear Dynamical Systems for Time-Series Data Analysis, *Stanford Computational Neuroscience Journal club*, 2019.

Projects

AR Hu, **GH Wilson**, *Persistent Homology of Global Functional Connectivity for Depression Classification*, 2017.

W Wang, L Huang, **GH Wilson**, J Price. Recurrent Neural Networks for Neoantigen Prediction, *SVAI Genomics Hackathon*, San Francisco 2017.

Teaching

Instructor, *Neurotechnology Research Review Decal*, UC Berkeley, Fall 2017.

Outreach

Volunteer, *Brain & Body NightLife*, California Academy of Sciences, November 2019.